

Practical Assignment Sheet (SQLite-Unit 1 & 2)

1. Create 'Sales' database.

2. Create tables and insert records:

2.1. Salesman (sid, name, city, commission) (primary key = SID)

Sid	Name	City	Commission
5001	Mohan Patel	Anand	0.15
5002	Nail Shah	Surat	0.13
5005	Preet Vyas	Ahmedabad	0.11
5006	Jeevan Mehta	Navsari	0.14
5003	Paul Adam	Nadiad	0.12
5007	Ramesh Patel	Surat	0.13

2.2. Order (oid, pur_amt, ord_date, sid)

Oid	Pur_amt	Ord_date	sid
1	12000	2020-12-05	5001
2	42000	2020-12-20	5002
3	2000	2021-02-02	5005
4	14000	2021-03-23	5005
5	23000	2021-04-15	5003
6	33000	2021-05-20	5001
7	32000	2021-06-22	5003
8	23500	2021-07-01	5003
9	43000	2021-07-05	5001
10	12000	2021-07-15	5002

2.3. Customer(cid,cust_name,grade,sid)

cid	Name	grade	sid
C1	Mohit Patel	100	5001
C2	Geeta Vyas	200	5003
C3	Jaya Patil	100	5005
C4	Vishal Gohel	300	5005
C5	Kartik Goyenka	200	5002
C6	Meera Prajapati	100	5001
C7	Veer Vyas	300	5002
C8	Maya Mehta	200	5003

3. Perform following tasks:

- 3.1. Display salesman's data with proper heading.
- 3.2. Display customer's data column-wise.
- 3.3. Display order's data in CSV format.
- 3.4. Display salesman's data in HTML format.
- 3.5. Display customer's data in Insert format.
- 3.6. Display all tables of SALES database.

- 3.7. Display structure of SALESMAN table.
- 3.8. Display all current settings.
- 3.9. Display current opened database.
4. Perform following tasks:
 - 4.1. Create transaction -> Insert two records in Customer table -> Apply rollback
 - 4.2. Create transaction -> Delete two records from Order table -> Apply rollback
 - 4.3. Create transaction -> Update name of '5003' to 'Raghav Panchal' -> Apply commit
5. Perform following tasks:
 - 5.1. Display customers' detail whose grade is 200.
 - 5.2. Display salesman's detail whose name start with 'P'.
 - 5.3. Display unique grades of customers.
 - 5.4. Display order's detail whose purchase amount is in between 10000 to 20000.
 - 5.5. Display customers' detail of SID 5001 and 5003.
 - 5.6. Display the top three order's detail.
 - 5.7. Display the top 5th, 6th and 7th orders' detail.
 - 5.8. Display Salesman's detail who doesn't have any customer.
 - 5.9. Display Salesman ID wise total purchase amount.
 - 5.10. Display Salesman ID wise detail whose average grade is more than 150.
 - 5.11. Display the Order detail in descending order as per Purchase amount.
 - 5.12. Display customers' detail with bonus. Calculate bonus as per following criteria:
 - Grade=100 -> Bonus = 5000
 - Grade=200 -> Bonus = 10000
 - Otherwise -> Bonus = 15000
 - 5.13. Display orders' detail with Class. Class is declared as per following criteria:
 - SID=5001 or 5002 -> Class='A'
 - SID=5003 or 5005 -> Class='B'
 - Otherwise -> Class='C'
 - 5.14. Display customers' detail with salesman name.
 - 5.15. Display orders' detail with salesman detail.
 - 5.16. Display orders' detail with salesman's commission.
6. Create trigger for following:
 - 6.1. Before inserting new record in Salesman table check whether commission is in between 0 or 1 or not. If not display error message.
 - 6.2. Check whether inserted or updated order_date in Order table is in 'yyyy-mm-dd' format or not. If not generate error message.
 - 6.3. Create one TRANSACTION_LOG table (ID, table_name, current_date) when new record is inserted into Salesman or Customer tables. (2 triggers)

Example: TRANSACTION_LOG

5008	Salesman	2021-07-10
11	Customer	2021-07-20

- 6.4. Create one AUDIT table (cid, old_name, new_name, old_sid, new_sid, operation, date) for Customer table's modified operations (For Update and delete). (2 triggers)
 - 6.5. If any record is deleted from Salesman table then first check whether related records are available in child table (customer). If available then generate error message and don't allow to delete record.
 - 6.6. Allow to modify the order's data between 10AM to 5PM only otherwise generate user defined error.
7. Dumping into files
 - 7.1. Dump entire 'SALES' database into file name 'sales.sql'.
 - 7.2. Dump Customer table's information into 'Customer.txt' file.
 - 7.3. Append Order table's information into same file 'Customer.txt'.
 - 7.4. Dump all tables' structures into 'Sales_schema.sql' file.
 - 7.5. Dump data of All tables into 'Sales_data.txt' file.
 8. Create following table named 'result'

Rno	Name	Gender	City	Percentage
101	Raj	Male	Surat	88
102	Umang	Male	Anand	60
103	Meera	Female	Bharuch	76
104	Maya	Female	Surat	80
106	Raghav	Male	Surat	85

- 8.1. Export result table as 'result.csv' file.
- 8.2. Import the 'result.csv' file as table named 'new_result'.